# When Do Echoes Help? Learned Echolocation for Exploring Unseen Buildings in a Sample-Accurate Acoustic Simulator

**Anonymous authors**
**Paper under double-blind review**

## Abstract

Bats and some blind humans find their way by emitting a sound and listening to its echoes. We ask when a simulated agent trained with reinforcement learning benefits from doing the same, and in what form the echoes should be presented to it. We first make an existing engine usable for this question. Steam Audio 4.8.1, a widely used ray-traced room-acoustics library, stores reflected energy in 10 ms bins and fills each bin with noise, so a wall 1 m away and a wall 2.5 m away produce the same impulse response. We patch its core to use one bin per audio sample and to emit one impulse per arrival; a reflection from a single wall then starts within one sample (21 $\mu$s at 48 kHz) of the image-source prediction, and 26 of 26 analytic checks pass (speed of sound fitted from arrivals: 340.01 m/s against the engine's 340 m/s). We connect the patched engine to Habitat-sim and train recurrent PPO agents that click once per step in complete darkness. On the Replica dataset (14 training buildings, 4 held-out buildings, 3 seeds per agent, 2 million steps each) we find: (i) when a goal vector is supplied, echoes do not improve navigation success in held-out buildings (0.38 with echoes, 0.39 without); (ii) when the agent must explore without any goal, echoes help substantially: an agent that receives a range-by-direction "sonar image" computed from the first-order Ambisonic echo response covers $37.0 \pm 1.5$ cells of 0.5 m in 300 steps, against $24.5 \pm 3.9$ for a blind agent with a bump sensor and 28.0 for a scripted bump-and-turn explorer, with fewer collisions than either; (iii) the same echoes presented as binaural spectrograms from three head orientations yield only $26.6 \pm 3.3$ cells at 2.7 times the simulation cost; (iv) when the echo carries distance but no direction, agents that choose where to aim each click do not cover more ground than a scripted sweep, but collide least of all agents ($15.4 \pm 2.0$ collisions per episode); raising the cost of a click from 0 to 0.1 lowers their click rate from 0.91 to 0.11 per step without lowering coverage, while collisions rise to $48.2 \pm 13.9$; (v) decoders trained on the agents' recurrent states recover which nearby floor cells the agent has already visited in held-out buildings (AUROC 0.76–0.83), and this path memory is strongest in the blind agents, while agents that hear the room hold local geometry in their current observation rather than in memory; (vi) erasing the recurrent memory every 20 steps reduces the blind agents' coverage by 15% and leaves the echo agents' coverage unchanged, and erasing it at every step multiplies the echo agents' collisions by 3 to 6 while they still cover 70–100% of their usual area. All code, the simulator patch, raw evaluation logs and figure scripts are released.

## 1 Introduction

A click reflected from a wall returns after $2d/c$ seconds, where $d$ is the distance to the wall and $c$ the speed of sound: 5.9 ms for a wall at 1 m. Echolocating bats use this delay, together with the direction from which

Disclosure required by TMLR: a large language model (an AI coding assistant) was used to write the software and to draft the text of this submission under the authors' direction; details are given in the statement "Use of AI assistance" before the references.

the echo arrives, to fly through clutter in darkness (Griffin, 1958; Moss & Surlykke, 2010), and some blind people use tongue clicks in the same way (Thaler & Goodale, 2016). An embodied agent that could do this would have a sense that works without light and needs only a loudspeaker and a few microphones.

Research on learned audio navigation has concentrated on a different problem: reaching a *sounding goal*, usually with vision available (Chen et al., 2020; Gan et al., 2020). Work on echoes has mostly treated them as a supervisory signal for visual representations (Gao et al., 2020) or as input for predicting depth maps (Christensen et al., 2020). The closest work, Zhu et al. (2022), reports a point-goal agent that receives its position and the binaural echoes of a sound it emits and reaches a higher SPL than an agent without echoes. To our knowledge, however, no study has compared different presentations of the echo signal to a policy, studied tasks without a goal vector, or reported results over several training seeds on explicitly held-out buildings. This paper does that.

Such a study needs a simulator in which the arrival time of each echo is correct to well under a millisecond, because 1 ms corresponds to 17 cm of range. We found that Steam Audio (Valve Corporation, 2026), a ray-traced acoustics engine that is open source, fast and actively maintained, does not provide this in its shipped form: it accumulates reflected energy in bins of 10 ms (3.4 m of path length) and reconstructs the impulse response by filling each bin with band-limited noise. Section 3 describes a small patch that removes this limit and its validation against analytic acoustics.

**Contributions.**

1. **A sample-accurate, validated echo simulator.** A 172-line patch to the Steam Audio 4.8.1 core (per-sample energy bins and a deterministic "sparse" reconstruction, both switched by environment variables), three fixes for defects in the upstream code, and a validation suite of 26 analytic checks on a single wall plus 7 checks in a shoebox room, all of which pass. We also document one physical deviation of the engine that the patch does not remove: specular reflection energy is too low by a factor $\cos\theta$ at incidence angle $\theta$ (Section 3.5).

2. **An environment for echo-based embodied learning.** A Gym environment that takes scene geometry, navigation mesh and agent motion from Habitat-sim (Savva et al., 2019) and computes the acoustics of every step with the patched engine on the same triangle mesh. It provides three echo observations of decreasing richness: binaural spectrograms from several head orientations, a range×azimuth *sonar image* obtained from the Ambisonic intensity vector of each arrival, and a direction-free *range profile* combined with a steerable click beam.

3. **A controlled study with three seeds per configuration on held-out buildings**, comprising four experiments: a comparison of echo representations for exploration (Experiment 1), a test of learned click aiming when the beam is the only source of direction (Experiment 2), a probe of what the agents' recurrent memory encodes (Experiment 3), and a causal test in which that memory is erased during an episode (Experiment 4). We also report, clearly marked as single-seed, the preliminary runs that shaped the design, including a negative result for goal-directed navigation.

**Summary of findings.** Echoes did not help our agents reach a goal whose direction and distance they were told at every step; a blind agent with a bump sensor did as well. Echoes helped when there was no goal vector and the agent had to cover unknown floor area: the sonar-image agent covered 51% more cells than the blind agent with 15% fewer collisions (Table 3). The representation mattered more than the amount of acoustic simulation: three binaural spectrograms per step were no better than blindness, one sonar image per step was clearly better. Giving the agent control over the click direction changed its behaviour (it became the most cautious agent and sometimes stopped clicking) but did not increase coverage. Finally, all agents carried a decodable record of their own recent path in memory, and the agents that could not sense walls relied on it most: erasing memory older than 20 steps reduced coverage only for the blind agents (Table 7).

## 2 Related work

**Audio in embodied AI.** SoundSpaces (Chen et al., 2020) introduced audio-goal navigation with precomputed room impulse responses on a discrete grid of Replica (Straub et al., 2019) and Matterport3D (Chang et al., 2017) scenes; SoundSpaces 2.0 (Chen et al., 2022) added continuous rendering with a bidirectional path tracer. Gan et al. (2020) studied a similar task. Follow-up work added waypoints and acoustic memory (Chen et al., 2021). In all of these the agent listens to an external source; the agent in our study is its own source. Our environment reuses the SoundSpaces action space and the 1 s source clips for the audio-goal sanity check, but computes acoustics with a different engine at every step.

**Echoes as a learning signal.** VisualEchoes (Gao et al., 2020) uses simulated echoes to pretrain image encoders; BatVision (Christensen et al., 2020) predicts depth images from recorded binaural chirp echoes; Purushwalkam et al. (2021) reconstruct floor plans from audio and vision. Robotic work has built bat-like ultrasonic platforms that map and navigate with hand-designed processing (Eliakim et al., 2018). Closer to our setting, Hu et al. (2023) train an agent with vision to decide when to emit a sound in order to map a building efficiently, and Yu et al. (2023) add a learned sound emitter to visual point-goal navigation. Zhu et al. (2022) train a point-goal agent in SoundSpaces whose inputs are its position and binaural echoes of its own emitted sweep, without vision, and report SPL 0.547 on Replica against 0.491 for a position-only agent; no seeds or variance are given and only one echo representation is used. Outside building-scale simulators, Mohan & Vanderelst (2020) and Nguyen & Vanderelst (2022) train echo-only deep reinforcement learning agents in two-dimensional bat simulations. To our knowledge no previous study combines policies driven by self-generated echoes without vision, a controlled comparison of echo representations, exploration without a goal vector, and evaluation over multiple seeds on held-out buildings.

**Exploration and memory in navigation agents.** Coverage rewards for learned exploration follow Chen et al. (2019) and Chaplot et al. (2020). The auxiliary loss we use is in the spirit of Mirowski et al. (2017) and Jaderberg et al. (2017). Our probing experiment follows the linear-probe methodology of Alain & Bengio (2016) and is closest to Wijmans et al. (2023), who showed that blind point-goal agents with only egomotion input build map-like representations. We extend that finding to exploration agents with acoustic and bump sensing and compare memory content across sensors.

**Acoustic simulation and spatial audio analysis.** The image-source model (Allen & Berkley, 1979) gives exact early reflections in a shoebox room and is our reference for validation. The direction estimate we use for the sonar image is the acoustic intensity vector of a first-order Ambisonic signal, as used in spatial impulse response rendering (Merimaa & Pulkki, 2005) and directional audio coding (Pulkki, 2007). Steam Audio traces rays with its own tracer or with Embree (Wald et al., 2014).

## 3 A sample-accurate echo simulator

### 3.1 What Steam Audio computes

We established the following from the Steam Audio 4.8.1 source code and confirmed each item by measurement with a C program that places one source, one listener and one planar wall in an otherwise empty scene (Appendix A).

Reflections are traced from the listener. At each surface hit the engine adds the irradiance arriving from the source, $1/(4\pi d^2)$ at source distance $d$, multiplied by a reflectance: a specular lobe $\frac{n+2}{8\pi}(\mathbf{h}\cdot\mathbf{N})^n$ with $n = 100$ weighted by $(1-s)$, plus a Lambertian term $\frac{s}{\pi}\cos\theta_i$, where $s$ is the material's scattering coefficient, $\mathbf{h}$ the half vector and $\mathbf{N}$ the surface normal; the result is multiplied by $(1-\alpha_b)$ for absorption $\alpha_b$ in each of three frequency bands (below 800 Hz, 800 Hz–8 kHz, above 8 kHz). The speed of sound is 340 m/s. Energy is accumulated in a histogram over time, band and spherical-harmonic channel (ACN order, N3D normalisation). Time zero of the histogram is the arrival of the direct sound, so a reflection of path length $r_{\text{refl}}$ appears at $(r_{\text{refl}} - r_{\text{direct}})/c$. Air absorption multiplies energy by $e^{-k_b d}$ with $k_b = 0.0002$, 0.0017, 0.0182 per metre. Triangles reflect from both sides. There is no diffraction and no phase change on reflection.

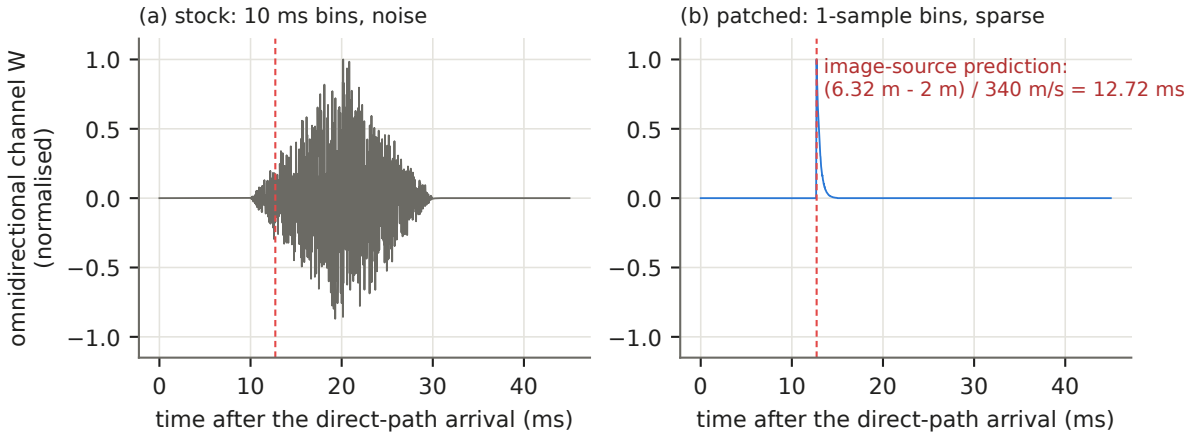



Figure 1: Impulse response (omnidirectional channel, 48 kHz) for a source 2 m from the listener and a fully reflecting wall 3 m to the side, so that the image source is 6.32 m away and the reflection should arrive 12.72 ms after the direct sound (dashed line). (a) Stock Steam Audio 4.8.1. (b) The patched core used in this paper; the short tail after the onset is the engine's glossy ($n = 100$) specular lobe.

### 3.2 The problem: 10 ms bins filled with noise

The histogram bins are 10 ms long. In 10 ms sound travels 3.4 m, so all surfaces within 1.7 m of range fall into the same bin. To turn the histogram into an impulse response the engine multiplies a fixed sequence of uniform noise in $[-1, 1]$ by a linear ramp between the square roots of consecutive bin energies and band-limits it. Figure 1a shows the result for one wall: the echo, which should be a single arrival at 12.72 ms, is a burst of noise extending from 10 to 30 ms. The total energy and the coarse timing are right (25 of 25 checks on the stock library pass once the binning is taken into account), but an agent cannot measure range from it.

### 3.3 The patch

The patch changes 11 files (172 lines added, 13 removed). With no environment variable set the library behaves exactly as shipped.

- **Bin duration.** The constant 0.01 s becomes a value read from `STEAMAUDIO_BIN_DURATION`. We set it to one sample period (1/24000 s in the learning experiments, 1/48000 s in the validation).
- **Sparse reconstruction.** With `STEAMAUDIO_RECONSTRUCTION=sparse`, each non-empty bin produces one impulse of amplitude $Y_m(\mathbf{d})\sqrt{4\pi E}$ in channel $m$, where $E$ is the bin's energy, $\mathbf{d}$ its energy-weighted direction and $Y_m$ the spherical harmonic. With this scale a mirror reflection of path length $r$ has energy $1/r^2$, consistent with the $1/r$ amplitude of the engine's direct path. Polarity is positive for a bin that is a local peak in time and pseudo-random otherwise, so dense late reverberation does not add coherently. The three frequency bands are recombined with a zero-phase, exactly complementary filter bank (low = squared low-pass magnitude, high = squared high-pass magnitude, mid = 1−low−high), so a frequency-independent reflection is reproduced as an exact impulse.
- **Embree path.** Steam Audio's Embree back end runs an ISPC kernel with its own hard-coded 10 ms constant; with fine bins every reflection landed in the first two bins. The patch passes the bin duration into the kernel. After this fix the Embree build reproduces the default tracer to four significant digits in onset, energy and direction, and all learning experiments use it (it is 1.9 times faster when several clicks are traced per step).
- **Upstream defects fixed.** (1) In builds without Intel IPP, the complex multiply-accumulate used an aligned SSE load on an unaligned pointer and crashed in multi-channel convolution. (2) The

Table 1: Principal validation results for the patched core in sparse mode (48 kHz). "Image source" denotes the analytic prediction for a mirror reflection.

| Quantity | Result |
|---|---|
| Echo onset, wall at 1–15 m | within 1 sample (21 $\mu$s) of $\lfloor (r_{\text{refl}} - 0.01\,\text{m} - r_{\text{direct}}) f_s / c \rfloor$ for 13 wall distances; the 1 cm is the engine's surface offset for ray hits |
| Speed of sound | 340.01 m/s fitted from the arrivals |
| Echo onset, 34–102 m paths | within 0.8 sample with $10^6$ rays (a 343 m/s medium would be 42–127 samples off) |
| Energy versus distance | $E\,r^2$ constant to 0.3% over 3–80 m; log–log slope $-2.003$ |
| Absolute level | measured/ideal $= 12.2/4\pi = 0.97$ |
| Absorption | $E(\alpha)/E(0) = 1 - \alpha$ to four decimals |
| Air absorption | per band within 0.018 of $e^{-k_b d}$ |
| Direction | first-order channels give the image-source direction; N3D gain 1.715 measured, $\sqrt{3} = 1.732$ expected |
| Shoebox, first-order echoes | all 6 begin at the image-source sample; energies within 15% of the image-source model including the $\cos\theta$ factor of Section 3.5 |
| Shoebox, reverberation | Schroeder $T_{60}$/Eyring $= 0.91$, 1.04, 1.11 for $\alpha = 0.1$, 0.3, 0.5 with diffuse walls |

compiler flag `-fabi-version=6` prevents compilation with GCC 10 and later. (3) The convolution effect and the Ambisonic decoder cross-fade from an empty state during the first audio frame they process, which attenuates anything arriving within that frame; our impulse-response extraction feeds one silent frame first.

### 3.4 Validation

Table 1 lists the principal checks; Appendix A gives all of them. The shoebox test (6×4×3 m room, 262,144 rays) compares against a three-bounce image-source model (Allen & Berkley, 1979) and against the Eyring reverberation formula.

### 3.5 Deviations from ideal acoustics that remain

**Grazing reflections are too weak.** With source and listener at equal height above a wall and separated laterally, the reflected energy times $r^2$ is 0.995, 0.962, 0.864, 0.707, 0.502, 0.346, 0.183 at incidence angles of 0, 15, 30, 45, 60, 70, 80 degrees, which equals $\cos\theta$ to within 0.5%. The specular lobe above lacks the $1/\cos\theta$ factor that makes a normalised half-vector lobe behave like a mirror. Onset times are unaffected. For an agent's own click, where source and listener coincide, the specular lobe returns energy only from surface points close to normal incidence (energy-weighted mean incidence angle 8.0°). We evaluated the lobe formula by ray casting on the meshes of the four held-out buildings at 50 random agent positions each: the first-bounce energy is 0.990 of that of a mirror-correct lobe on average (minimum over the 200 positions 0.976), and 0.989 (minimum 0.952) for the returns of the first 2 ms. First echoes therefore lose about 1% of their energy; paths with several bounces include oblique reflections and are attenuated more. **Glossy lobe.** The $n = 100$ lobe spreads each echo over an exponential tail whose mean excess path is $2w/(n+1)$ for a wall at distance $w$ (6 cm at 3 m). **Not modelled:** diffraction, reflection phase, and frequency resolution finer than three bands.

## 4 Environment, observations and agents

### 4.1 Embodiment and acoustics

The agent is a cylinder of radius 0.1 m with its ears and mouth at 1.5 m height. Actions follow SoundSpaces: *stop*, *move forward* 0.25 m, *turn left* 10°, *turn right* 10°. Unlike SoundSpaces we disable Habitat's wall sliding: a forward move that would enter an obstacle leaves the agent where it is and is reported as a collision. The agent receives no image at any time.

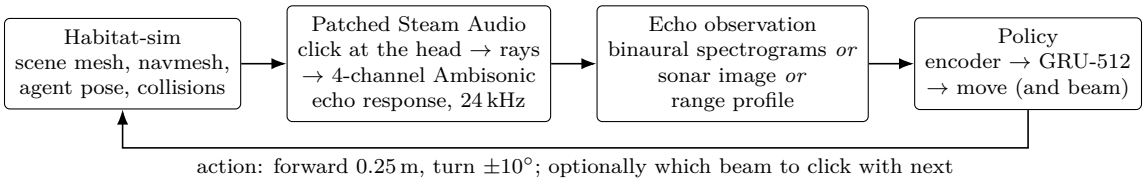



Figure 2: One environment step. Habitat-sim moves the agent; Steam Audio traces the agent's click on the same triangle mesh; the echo response is converted to one of three observations; a recurrent policy chooses the next movement and, in some configurations, the beam of the next click.

At every step a click is emitted at the head position and the patched engine traces its reflections on the scene's full triangle mesh (for Replica, the `mesh_semantic.ply` files, rotated into Habitat's coordinate frame). Settings: 2,048 rays, 4 bounces, 0.3 s response, first-order Ambisonics (4 channels), 24 kHz, one Steam Audio thread per environment, air absorption on. Every surface has the same material: absorption $(0.10, 0.20, 0.30)$ in the three bands and scattering 0.05. Because source and listener coincide, time zero is the instant of emission and a surface at distance $d$ returns at $2d/c$. The click's own direct sound is not part of the observation.

### 4.2 Three echo observations

**Binaural head scan.** The click is emitted three times, with the head (and a click directivity $((1+\cos\gamma)/2)^2$ around the facing direction, $\gamma$ being the angle from it) turned $-45°$, $0°$ and $+45°$ relative to the body. Each Ambisonic response is decoded to two ears with Steam Audio's default head-related transfer function, resampled to 16 kHz, and converted to a log-magnitude spectrogram of its first 128 ms (FFT length 128, hop 16 samples, then averaged over pairs of frames), giving 65 frequency bins × 64 time frames of 2 ms × 2 ears per orientation, 6 channels in total. This is the observation closest to what a human echolocator hears.

**Sonar image.** One omnidirectional click per step. In the sparse reconstruction each sample $t$ of the response is one arrival. Its omnidirectional channel $W(t)$ gives the amplitude and its three first-order channels, rearranged to world axes, equal $\sqrt{3}\,W(t)\,\mathbf{u}(t)$, where $\mathbf{u}(t)$ is the unit vector pointing towards the direction of arrival. With the agent's forward, right and up unit vectors $\mathbf{f}, \mathbf{r}, \mathbf{v}$,

$$\varphi(t) = \operatorname{atan2}\big(-\mathbf{u}\cdot\mathbf{r},\ \mathbf{u}\cdot\mathbf{f}\big), \qquad \sin\varepsilon(t) = \mathbf{u}\cdot\mathbf{v}, \tag{1}$$

are the azimuth (positive to the left) and elevation of the arrival in the body frame. We keep the first 128 ms (3,072 samples) and accumulate arrivals into a grid of 64 range bins (48 samples each, i.e. 2 ms of delay or 17 cm of range, up to 10.9 m) by 32 azimuth bins (11.25° each, covering the full circle, with "ahead" in the centre column):

$$I_0[i,j] = \log\Big(1 + 10 \sum_{t\in\mathrm{bin}(i,j)} |W(t)|\cos\varepsilon(t)\Big), \qquad I_1[i,j] = \operatorname{sgn}(S)\log(1+10\,|S|), \quad S = \sum_{t\in\mathrm{bin}(i,j)} |W(t)|\sin\varepsilon(t). \tag{2}$$

Channel $I_0$ is a top-down polar map of horizontal returns; channel $I_1$ collects returns from the floor (negative) and ceiling (positive) so that they do not clutter $I_0$. The observation has shape $64 \times 32 \times 2$. Figure 3b shows $I_0$ in a shoebox room: the four walls appear at their true distances (errors 3–11 cm, at most one range bin) and in the correct azimuth columns. No head-related transfer function is involved, which is why this observation needs one traced click instead of three and no binaural decoding.

**Range profile and click beams.** To study aiming we remove direction from the echo altogether: the range profile is $P[i] = \log(1 + 10\sum_{t\in\mathrm{bin}(i)} |W(t)|)$ over the same 64 range bins, a vector of 64 numbers, as a single-microphone ultrasonic range finder would deliver. Direction can then enter only through the click's emission pattern. Steam Audio's source directivity is $|(1-w) + w\cos\gamma|^p$ for dipole weight $w$ and power $p$. We use $w = 0.5$, $p = 8$, i.e. $((1+\cos\gamma)/2)^8$, a forward lobe that falls to half its peak value 33° off axis, to

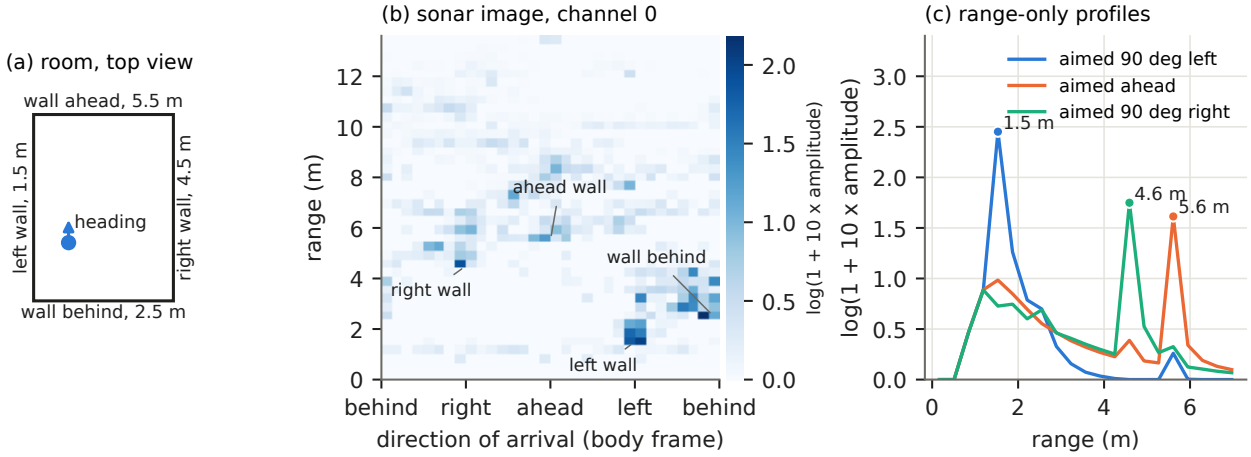



Figure 3: The echo observations in a 6×8×3 m shoebox room with the simulation settings of the learning experiments. (a) Agent position and heading. (b) Sonar image channel $I_0$ for one omnidirectional click (4 bounces); the nearest point of each wall appears at its distance and bearing, later entries are multi-bounce paths. (c) Range profiles for a forward-lobe click aimed left, ahead and right (1 bounce): the peak moves to 1.5, 5.6 and 4.6 m, the distances of the left, front and right walls (1.5, 5.5 and 4.5 m).

0.4% at 90° and to zero directly backwards, aimed at one of five yaw angles relative to the body: 90° left, 45° left, ahead, 45° right, 90° right. Figure 3c confirms that the strongest return of each aimed beam is the wall it points at. (In Experiment 1 the "learned beam" agent used an earlier beam set with $w = 1$, $p = 4$; because of the absolute value this is a figure-of-eight, $|\cos\gamma|^4$, that fires equally backwards. We report those results as run and note this where relevant.)

### 4.3 Tasks and rewards

**Exploration in the dark (main experiments).** An episode lasts exactly 300 steps and starts at a random navigable position with a random heading. The floor is divided into square cells of 0.5 m; the agent receives +1 the first time it enters a cell, −0.01 every step, and $-\lambda$ for every step on which it collides ($\lambda = 0.05$ in Experiments 1 and 2). *Stop* is treated as *move forward*. The two reported quantities are **coverage**, the number of distinct cells entered in an episode (the start cell counts, so the minimum is 1), and **collisions per episode**. In 300 steps an agent can travel at most 75 m.

**Point-goal in the dark (preliminary experiments).** The agent is given the straight-line distance and bearing to a goal at every step and must issue *stop* within 1 m of it; episodes are limited to 200 steps and start 1–8 m (geodesic) from the goal. Reward is the decrease in geodesic distance, −0.01 per step, +10 for success and $-\lambda$ per collision. Metrics are success rate and SPL (Anderson et al., 2018).

**Bump sensor.** In exploration every agent, including the blind one, observes one bit: whether its previous action collided.

### 4.4 Agents and training

Each image-like observation is encoded by a three-layer convolutional network (32, 64, 32 channels; kernels $8 \times 4$, $4 \times 3$, $3 \times 3$; strides $(4, 2)$, $(2, 2)$, 1) followed by a linear layer to 512 units; the range profile by a two-layer perceptron (256, 256); the bump bit by a 16-unit layer. These features are concatenated with a 32-dimensional embedding of the previous action and fed to a GRU (Cho et al., 2014) with 512 units, followed by linear policy and value heads. Agents with a learned beam have a second policy head over beams (sampled independently of the movement) and receive a 16-dimensional embedding of the beam used for the

current observation. Parameter counts: 2.09 M (sonar), 2.12 M (sonar with beam head), 2.62 M (head scan), 1.41 M (range), 0.93 M (blind).

Training uses PPO (Schulman et al., 2017) with generalised advantage estimation (Schulman et al., 2016): 14 parallel environments, one per training building; rollouts of 128 steps; 4 epochs over 4 minibatches of whole environment sequences; Adam with learning rate $2.5 \times 10^{-4}$; $\gamma = 0.99$, $\lambda_{\mathrm{GAE}} = 0.95$, clipping 0.2, entropy coefficient 0.02, value coefficient 0.5, gradient-norm clipping 0.5; 1,117 updates, i.e. 2,001,664 environment steps.

**Auxiliary loss.** All echo agents additionally predict, from the GRU state, $\log(1 + d_k)$ for the distance $d_k$ (capped at 10 m) to the nearest surface along five horizontal directions ($-90°, -45°, 0°, 45°, 90°$ from the heading), with a mean-squared-error loss weighted 0.5. The distances come from ray casts during training and are never observed. The blind agents in Experiment 1 were trained without this loss.

**Blank-echo augmentation.** In Experiments 1 and 2 the echo observation of each training step is replaced by zeros with probability 0.05, so that a missing echo is not out of distribution.

### 4.5 Buildings and evaluation protocol

We use the 18 scenes of Replica (Straub et al., 2019). Four are held out and never seen during training: `office_2`, `frl_apartment_4`, `hotel_0`, `room_2` (navigable area 12.5, 39.6, 16.0 and 13.2 $\mathrm{m}^2$). The other 14 are used for training (5.4–89.0 $\mathrm{m}^2$; Appendix B).

Every trained agent is evaluated on 30 episodes in each held-out building. The 30 start poses of a building are identical for all agents. Unless stated otherwise we report the **greedy** policy (most probable action); Appendix C also gives results for sampled actions. For each seed we average over the 120 episodes; we then report the mean and the population standard deviation over the three seeds, written mean $\pm$ s.d. With three seeds these standard deviations are rough, so we also state per-seed values where a conclusion depends on them.

**Baselines.** *Bump-and-turn*: move forward until a collision, then turn left by 30–90° (uniformly chosen multiple of 10°). *Random*: uniformly random actions.

**Echo ablations.** On `office_2` and `hotel_0` each echo agent is re-evaluated with its echo observation replaced (a) by zeros and (b) by an echo observation recorded at a random other pose in the same building ("shuffled"). If an agent's behaviour does not change under (b), it is not using the content of its echoes.

## 5 Preliminary experiments that shaped the design (single seed)

The runs in this section were made with one seed per configuration while the system was being developed. They motivated the main experiments but, as Section 6 shows, single runs differ by several cells and by a factor of two to three in collisions, so they should be read as observations, not as established effects.

**P1. Sanity check: reaching a sounding goal.** With SoundSpaces' task definition (a telephone rings at the goal, wall sliding on) in one Habitat test apartment, a PPO agent trained for 1 million steps on binaural spectrograms from our simulator reached the goal in 75% (greedy) and 79% (sampled) of 100 unseen start–goal pairs, with SPL 0.67; a random agent reached 2%. The simulated binaural cues are therefore learnable in the usual way.

**P2. Echoes are ignored unless collisions cost something.** For point-goal in the dark in the same apartment, an agent trained with $\lambda = 0$ behaved identically with real, shuffled or zeroed echoes. With $\lambda = 0.1$, zeroing the echoes at test time multiplied collisions by 3 to 22 while leaving success unchanged, in two seeds of two echo agents (Appendix E). All later runs use a collision penalty.

Table 2: Preliminary result P4 (one seed per agent): point-goal navigation in the dark on the four held-out Replica buildings, greedy policy, 100 episodes per building. Entries are success rate / SPL / collisions per episode. Random actions succeed in 0.01–0.12 of episodes.

| Held-out building | Head-scan echo + auxiliary loss | Blind |
|---|---|---|
| office_2 | 0.34 / 0.31 / 6.1 | 0.39 / 0.38 / 3.8 |
| frl_apartment_4 | 0.43 / 0.38 / 6.9 | 0.28 / 0.28 / 3.0 |
| hotel_0 | 0.37 / 0.30 / 5.2 | 0.45 / 0.44 / 3.0 |
| room_2 | 0.39 / 0.35 / 4.2 | 0.44 / 0.43 / 4.0 |
| Mean | 0.38 / 0.34 / 5.6 | 0.39 / 0.38 / 3.5 |

Table 3: Experiment 1. Exploration in four held-out Replica buildings (300 steps, greedy policy, 30 episodes per building; mean ± s.d. over 3 seeds). Coverage is the number of distinct 0.5 m cells entered. "Sim. speed" is training throughput with 14 environments on one 8-core CPU and "Train time" the wall-clock time for 2 million steps.

| Agent | Coverage (cells) | Collisions / episode | Path (m) | Sim. speed (steps/s) | Train time (h) |
|---|---|---|---|---|---|
| Sonar image, fixed click | **37.0 ± 1.5** | 17.5 ± 6.7 | 31.0 | 490 | 1.13 |
| Sonar image + learned beam | 33.6 ± 4.7 | **13.2 ± 2.4** | 25.8 | 495 | 1.12 |
| Head-scan binaural | 26.6 ± 3.3 | 17.4 ± 3.6 | 18.0 | 181 | 3.08 |
| Blind (bump sensor) | 24.5 ± 3.9 | 20.6 ± 5.3 | 20.5 | 1016 | 0.55 |
| Scripted bump-and-turn | 28.0 | 31.8 | 22.2 | – | – |
| Random actions | 7.7 | 139.0 | 3.1 | – | – |

**P3. Three training buildings do not transfer.** Head-scan agents trained on three buildings for 2 million steps beat blind agents in those buildings (greedy point-goal success 0.88 against 0.73 in the largest) but fell to the blind level in an unseen Matterport3D house (0.18 against 0.21), and shuffling their echoes there changed nothing. This motivated the move to 14 training buildings.

**P4. With a goal vector, echoes do not improve success in held-out buildings.** Table 2 compares a head-scan echo agent (with auxiliary loss) and a blind agent, both trained for 2 million steps on the 14 Replica training buildings with $\lambda = 0.1$. Mean held-out success is 0.38 with echoes and 0.39 without; the blind agent has higher SPL and fewer collisions. Zeroing the echo agent's echoes on `office_2` and `hotel_0` did not lower its success (0.44 and 0.38) but raised its collisions from 6.1 and 5.2 to 9.6 and 11.0 per episode. The echo agent thus used its echoes only to avoid walls, something the blind agent achieved equally well from the bump bit and the goal vector. This single-seed observation differs from Zhu et al. (2022), who report a gain from echoes for point-goal navigation (with SoundSpaces acoustics and their own training setup); we did not investigate the discrepancy further and turned to a task without a goal vector.

**P5. The collision penalty sets a coverage–collision trade-off.** For exploration with the sonar image and a fixed click, $\lambda = 0.02$ gave 41.9 cells and 38.6 collisions per episode on the held-out buildings (one seed); $\lambda = 0.05$ gave $37.0 \pm 1.5$ cells and $17.5 \pm 6.7$ collisions (three seeds, Experiment 1). With $\lambda = 0.1$ a head-scan explorer became progressively more hesitant during training: its checkpoint at 0.5 million steps covered 26.5 and 21.3 cells on `office_2` and `hotel_0`, its final policy 12.4 and 17.1. We fixed $\lambda = 0.05$ for the main experiments.

## 6 Experiment 1: which echo representation helps exploration?

**Setup.** Four agents, three seeds each, exploration in the dark with $\lambda = 0.05$: (1) *sonar image, fixed click*: one omnidirectional click per step; (2) *sonar image + learned beam*: the agent chooses each step among no click, an omnidirectional click, and a $|\cos\gamma|^4$ beam centred 45° left, ahead or 45° right, and pays 0.01 reward per click; (3) *head-scan binaural*; (4) *blind*: bump bit only.

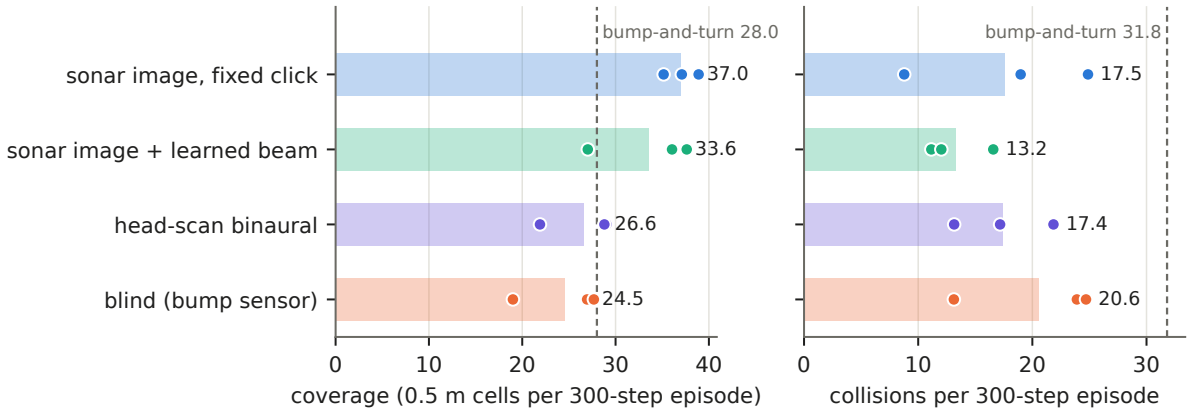



Figure 4: Experiment 1, per-seed results behind Table 3. Each dot is one training seed (mean over 120 held-out episodes); bars and numbers are means over seeds; the dashed line is the scripted bump-and-turn explorer.

**Result 1: the sonar image clearly improves exploration over blindness.** The sonar agent with a fixed click covers $37.0 \pm 1.5$ cells against $24.5 \pm 3.9$ for the blind agent, an increase of 12.5 cells (51%). The three sonar seeds (38.9, 35.2, 37.1) all exceed the three blind seeds (19.0, 27.0, 27.7). It also exceeds the scripted explorer (28.0) in every held-out building (Appendix C) while colliding about half as often (17.5 against 31.8). In terms of floor area, the sonar agent visits on average 51%, 35%, 42% and 76% of the navigable area of `office_2`, `frl_apartment_4`, `hotel_0` and `room_2` within 300 steps (25.6, 55.8, 26.9 and 39.9 cells of $0.25\,\mathrm{m}^2$).

**Result 2: the binaural head scan is not better than blindness.** The head-scan agent covers $26.6 \pm 3.3$ cells, within one standard deviation of the blind agent, although each of its steps traces three clicks and decodes three binaural responses, which makes its training 2.7 times slower than the sonar agent's. The underlying echo responses are the same in both cases; what differs is that the sonar image hands the policy range and bearing explicitly, whereas the spectrogram leaves the policy to recover them from interaural differences after a head-related transfer function.

**Result 3: beam choice in the sonar agent is within seed noise.** The learned-beam variant covers $33.6 \pm 4.7$ cells with $13.2 \pm 2.4$ collisions; both differences from the fixed click are smaller than the spread across seeds. The three seeds adopt different habits: clicking on 63%, 100% and 85% of steps and preferring different beams. This is unsurprising, because the sonar image already resolves direction, so the beam only re-weights it; and, as noted in Section 4.2, this beam set also fired backwards. Experiment 2 tests aiming in a setting where it matters.

**Do the agents use the content of their echoes?** Table 4 shows the ablations. Both sonar agents lose coverage and gain collisions when echoes are shuffled (fixed click: 26.2 → 19.8 cells and 21.1 → 31.8 collisions), so they read geometry from the image. With zeroed echoes the fixed-click sonar agent slows down sharply (11.1 cells, 8.7 collisions): despite the 5% blank-echo augmentation it does not move confidently without echoes. The head-scan agent is essentially unaffected by either ablation (17.4 cells with real echoes, 17.6 shuffled, 18.5 zeroed), which is consistent with Result 2: on these buildings it does not rely on its echoes.

**Learning curves.** Figure 5 shows training coverage and collisions. The blind agents plateau near 33 cells after 0.5 million steps; the sonar agents keep improving to 47 cells at 2 million steps and had not converged. Training coverage is higher than held-out coverage for all agents because the training set contains larger buildings.

Table 4: Echo ablations on `office_2` and `hotel_0` (greedy, mean ± s.d. over 3 seeds). Entries are coverage in cells / collisions per episode. "Shuffled" replaces each echo observation by one recorded at a random other pose of the same building.

| Agent | Real echoes | Shuffled echoes | Zeroed echoes |
|---|---|---|---|
| Sonar image, fixed click | $26.2 \pm 0.5$ / $21.1 \pm 8.8$ | $19.8 \pm 1.2$ / $31.8 \pm 8.2$ | $11.1 \pm 4.3$ / $8.7 \pm 2.7$ |
| Sonar image + learned beam | $23.4 \pm 3.1$ / $15.3 \pm 4.3$ | $14.9 \pm 4.3$ / $26.0 \pm 12.3$ | $15.0 \pm 9.4$ / $19.3 \pm 17.4$ |
| Head-scan binaural | $17.4 \pm 2.7$ / $19.3 \pm 6.9$ | $17.6 \pm 1.4$ / $21.5 \pm 3.4$ | $18.5 \pm 2.8$ / $18.3 \pm 6.6$ |
| Range + learned beam | $22.1 \pm 1.0$ / $16.7 \pm 3.5$ | $19.9 \pm 5.6$ / $25.2 \pm 9.9$ | $15.6 \pm 3.0$ / $13.1 \pm 3.9$ |
| Range + scripted sweep | $24.6 \pm 0.5$ / $25.6 \pm 3.1$ | $22.7 \pm 0.3$ / $34.1 \pm 4.2$ | $16.6 \pm 6.7$ / $31.3 \pm 27.6$ |
| Range + fixed ahead beam | $21.6 \pm 3.8$ / $28.4 \pm 8.1$ | $20.4 \pm 1.8$ / $29.8 \pm 9.3$ | $18.9 \pm 9.6$ / $25.2 \pm 17.5$ |
| Range + omni click | $21.5 \pm 5.6$ / $30.9 \pm 16.3$ | $20.5 \pm 5.1$ / $33.9 \pm 17.3$ | $20.8 \pm 1.2$ / $65.3 \pm 7.7$ |

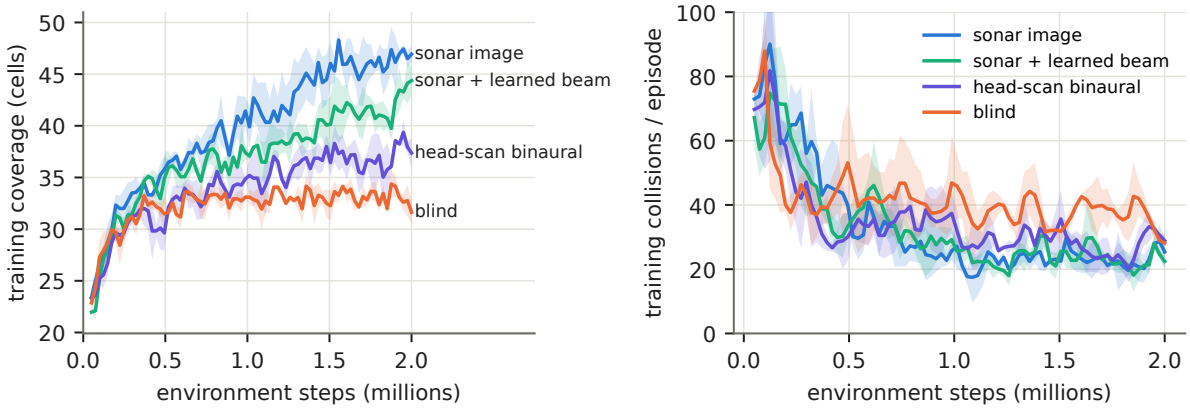



Figure 5: Experiment 1, training curves on the 14 training buildings (mean over 3 seeds; shading is ±1 s.d.). Left: coverage over the last 100 training episodes. Right: collisions per episode.

## 7 Experiment 2: does learned click aiming help when the beam is the only source of direction?

**Setup.** All four agents observe the 64-number range profile and the bump bit and are trained exactly as in Experiment 1 (including the auxiliary loss and blank-echo augmentation). They differ only in how the click is emitted: (1) *learned beam*: each step the agent chooses one of the five forward beams of Section 4.2 or no click, at a cost of 0.01 per click; (2) *scripted sweep*: the beam cycles through the five directions, one per step, in fixed order (the agent is not told the current direction and must track the cycle in memory); (3) *fixed ahead beam*; (4) *omnidirectional click*.

**Result 4: direction in the echo is worth 5 to 9 cells.** Every range-only agent lies between the blind agent (24.5 cells) and the sonar-image agent (37.0 cells). The best range-only mean, 32.0 cells, is 5.0 below the sonar image; the learned-beam agent is 9.3 below.

**Result 5: learned aiming does not increase coverage, but produces the safest explorer.** The learned-beam agent covers $27.7 \pm 1.4$ cells, fewer than the fixed ahead beam ($32.0 \pm 3.2$) and the scripted sweep ($30.4 \pm 0.3$). Its collisions are the lowest of any agent in this paper with the smallest spread, $15.4 \pm 2.0$ per episode (seeds: 13.0, 18.0, 15.2), and its path is the shortest (17.9 m): it trades distance travelled for not hitting things. By contrast the fixed-beam and omnidirectional agents vary between 9 and 41 collisions per episode across seeds (13.6, 33.6, 40.8 and 38.8, 40.2, 9.0).

**Result 6: agents ration clicks to different degrees.** With a click cost equal to the per-step time cost (0.01), seed 1 of the learned-beam agent stays silent on 62% of steps (clicks on 21–46% of steps depending

Table 5: Experiment 2. Range-only echo observation with four ways of emitting the click; same protocol as Table 3 (mean ± s.d. over 3 seeds). The first and last rows repeat Experiment 1 for reference.

| Agent | Coverage (cells) | Collisions / episode | Path (m) | Clicks per step (seeds 1, 2, 3) |
|---|---|---|---|---|
| Sonar image, fixed click (Exp. 1) | 37.0 ± 1.5 | 17.5 ± 6.7 | 31.0 | 1 |
| Range + fixed ahead beam | 32.0 ± 3.2 | 29.3 ± 11.5 | 24.4 | 1 |
| Range + scripted sweep | 30.4 ± 0.3 | 24.4 ± 3.3 | 23.5 | 1 |
| Range + omni click | 28.5 ± 8.1 | 29.3 ± 14.4 | 20.3 | 1 |
| Range + learned beam | 27.7 ± 1.4 | **15.4 ± 2.0** | 17.9 | 0.38, 0.96, 0.76 |
| Blind (Exp. 1) | 24.5 ± 3.9 | 20.6 ± 5.3 | 20.5 | 0 |

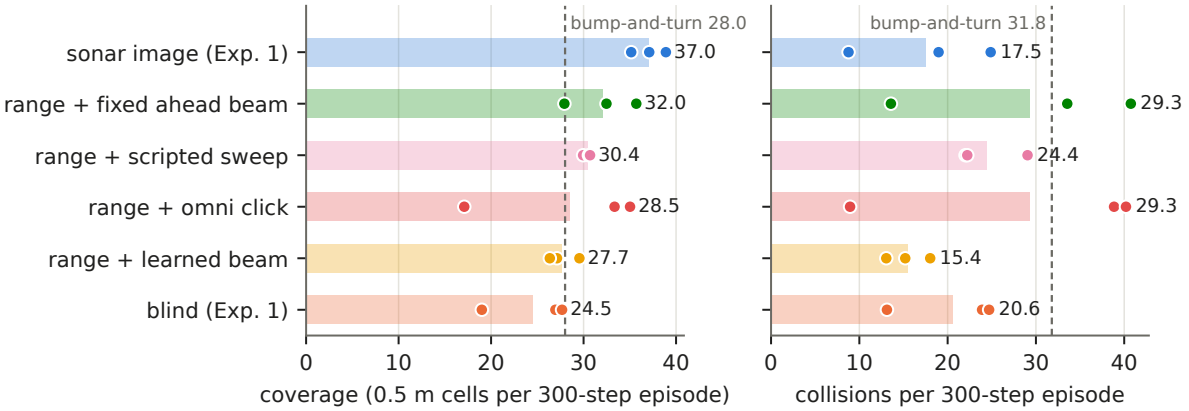



Figure 6: Experiment 2, per-seed results behind Table 5. Dots are seeds; bars and numbers are means; the dashed line is the scripted bump-and-turn explorer.

on the building), seed 3 on 24%, and seed 2 clicks on 96% of steps. Preferred beams differ by seed: seed 1 clicks mostly 45° left (14–20% of steps, depending on the building) and 90° right (3–17%), seed 2 mostly ahead (33–54%), seed 3 mostly 45° left (22–46%). Zeroing the range profile reduces the learned-beam agents' coverage from 22.1 to 15.6 cells on the two ablation buildings (Table 4), so the echoes obtained with the chosen beams are used.

**Result 7: a scripted sweep is the most reproducible range-only sensor.** Its three seeds cover 30.0, 30.6 and 30.7 cells. Sweeping gives the recurrent policy five directions every five steps without any learning of where to look.

**What does a click buy? A sweep of the click cost.** We retrained the learned-beam agent with the cost per click set to 0, 0.03 and 0.1 (three seeds each; 0.01 is the setting above; the per-step time cost is 0.01 and a new cell is worth 1). Table 6 shows that the click rate falls steadily with the cost, from 0.91 clicks per step when clicks are free to 0.11 at a cost of 0.1, where one seed never clicks at all. Coverage does not fall with it: it stays between 26.8 and 31.9 cells, within the spread across seeds. Collisions do change: with clicks at a cost of 0.1 the agents collide 48.2 ± 13.9 times per episode (the seed that never clicks: 62.9), about three times as often as at a cost of 0.01, and more than twice as often as the blind agents of Experiment 1 (20.6 ± 5.3). In this task, therefore, the aimed range clicks are used to avoid collisions, not to find new floor; an agent that has learned to rely on them and then gives them up for cost reasons moves as far but hits much more. Even when clicks are free the agents do not click on every step (0.81, 0.95 and 0.96 clicks per step).

Table 6: Click-cost sweep for the range + learned-beam agent (held-out buildings, greedy policy, mean ± s.d. over 3 seeds).

| Cost per click | Coverage (cells) | Collisions / episode | Clicks per step (seeds 1, 2, 3) |
|---|---|---|---|
| 0 | $29.4 \pm 2.6$ | $20.3 \pm 4.4$ | 0.81, 0.95, 0.96 |
| 0.01 (Table 5) | $27.7 \pm 1.4$ | $15.4 \pm 2.0$ | 0.38, 0.96, 0.76 |
| 0.03 | $26.8 \pm 5.8$ | $17.1 \pm 6.6$ | 0.95, 0.34, 0.38 |
| 0.1 | $31.9 \pm 0.9$ | $48.2 \pm 13.9$ | 0.19, 0.13, 0.00 |

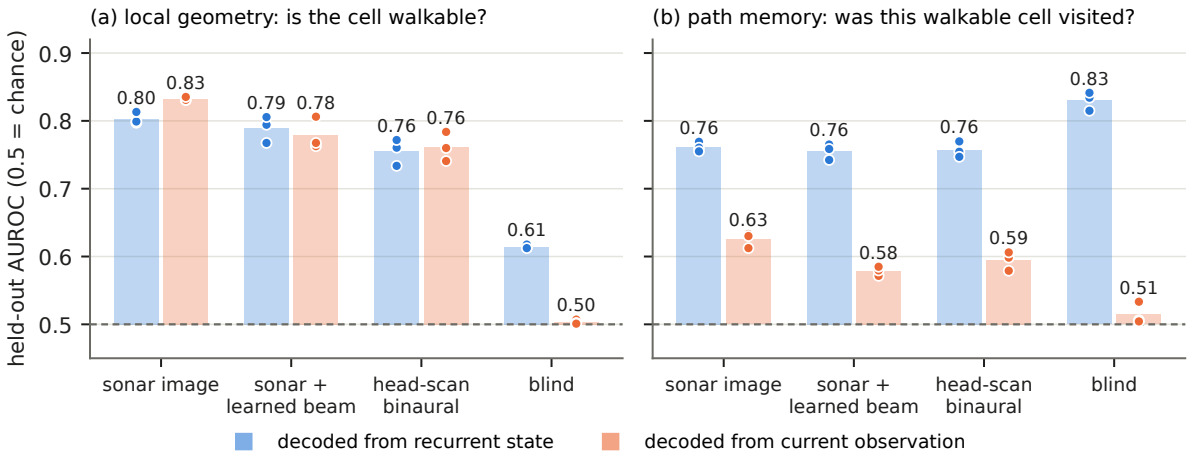



Figure 7: Experiment 3. Held-out AUROC of decoders that read the agent's recurrent state (blue) or only its current observation (orange). Dots are the three seeds of each agent, bars and numbers their mean. (a) Which of the 120 surrounding cells are walkable. (b) Which walkable cells the agent has already visited in this episode.

## 8 Experiment 3: what do the agents remember?

**Setup.** We take the 12 trained agents of Experiment 1 and roll each out with sampled actions, without further training, for 3,000 steps in each of the 14 training buildings and 6,000 steps in each of the 4 held-out buildings. At every step we store the 512-dimensional GRU state $h_t$, a reduced copy $x_t$ of the current observation (the echo observation average-pooled to at most 512 numbers, plus the bump bit; for the blind agent $x_t$ is the bump bit alone), and two binary labels for each cell of an $11 \times 11$ grid of 0.5 m cells centred on the agent and aligned with its heading (5.5 m × 5.5 m): *walkable*, whether the cell centre lies on the navigation mesh, and *visited*, whether the agent has entered that cell earlier in the episode. This yields 41,860 training and 23,920 held-out samples per agent. On average 29% of cells are walkable and 11% visited.

For each agent and each label we train a decoder (one hidden layer of 256 units, sigmoid outputs for the 120 non-centre cells; Adam, learning rate $10^{-3}$, weight decay $10^{-4}$, 15 epochs, batch size 512, standardised inputs) on the samples from the training buildings and report the area under the ROC curve (AUROC) on the held-out buildings, averaged over cells; 0.5 is chance. We do this once with $h_t$ as input and once with $x_t$. Linear decoders give the same ordering (Appendix D).

Visited cells are always walkable, so a decoder could score above chance on *visited* merely by recognising walls. We therefore evaluate *visited* only over the cells that are walkable in the sample at hand ("path memory"): of the floor that the agent could have stepped on, which part has it actually stepped on? No single echo contains this information.

**Result 8: all agents carry a record of their own path, readable in unseen buildings.** From the recurrent state, path memory is decoded with AUROC $0.762 \pm 0.006$ (sonar, fixed click), $0.755 \pm 0.010$

Table 7: Experiment 4. Exploration on `office_2` and `hotel_0` with the recurrent memory erased every 20 steps or every step (greedy, mean ± s.d. over 3 seeds). Entries: coverage in cells / collisions per episode / path length in metres.

| Agent | Memory intact | Erased every 20 steps | Erased every step |
|---|---|---|---|
| Blind | $21.3 \pm 3.2$ / $22.2 \pm 6.1$ / 18.9 | $18.2 \pm 2.4$ / $17.6 \pm 2.1$ / 14.6 | $3.2 \pm 0.0$ / $1.0 \pm 0.0$ / 1.0 |
| Sonar image, fixed click | $25.4 \pm 0.8$ / $22.7 \pm 9.5$ / 23.6 | $27.5 \pm 1.0$ / $20.8 \pm 8.6$ / 24.6 | $20.9 \pm 2.4$ / $71.4 \pm 6.9$ / 14.7 |
| Sonar image + learned beam | $23.3 \pm 3.7$ / $15.1 \pm 4.3$ / 20.9 | $23.9 \pm 1.8$ / $17.9 \pm 4.3$ / 19.8 | $16.3 \pm 2.8$ / $89.3 \pm 15.5$ / 12.6 |
| Range + learned beam | $21.8 \pm 1.5$ / $16.9 \pm 3.2$ / 15.1 | $24.2 \pm 1.8$ / $18.1 \pm 2.8$ / 15.5 | $22.8 \pm 1.8$ / $78.6 \pm 8.0$ / 14.5 |

(sonar, learned beam), $0.757 \pm 0.009$ (head scan) and $0.830 \pm 0.011$ (blind). From the current observation it is $0.625 \pm 0.009$, $0.579 \pm 0.005$, $0.594 \pm 0.011$ and $0.515 \pm 0.013$. The agents were trained only with the coverage reward (and, for echo agents, the wall-distance loss); nothing asked them to store where they had been.

**Result 9: the blind agents remember their path best.** Their path-memory AUROC of 0.830 exceeds that of every echo agent ($\leq 0.762$) in all three seeds. An agent that cannot sense walls can avoid re-covering ground only by integrating its own movements; an agent that hears the room has other cues for where open, unvisited space lies.

**Result 10: echo agents perceive local geometry rather than remember it.** For walkable cells, the sonar agent's current observation decodes better than its memory ($0.832 \pm 0.002$ against $0.803 \pm 0.007$); for the learned-beam and head-scan agents the two are equal within noise (0.779 against 0.789; 0.761 against 0.755). The recurrent state thus retains somewhat less geometric information than a pooled copy of the latest echo contains. The blind agents' memory holds a weak map of walkable space built from bumps and movement alone ($0.614 \pm 0.002$; the bump bit by itself gives 0.503).

## 9 Experiment 4: erasing memory during an episode

Experiment 3 shows what can be read out of memory, not whether the agents need it. Here we remove it.

**Setup.** We re-evaluate trained agents on `office_2` and `hotel_0` (greedy policy, the same 30 start poses per building as before, 3 seeds) while setting the GRU state to zero every $N$ steps. The zero state is the state every training episode starts with, so the erased state itself is familiar to the policy. The embedding of the previous action (and beam) is still supplied. $N = 20$ removes everything older than at most 20 steps (at most 5 m of travel); $N = 1$ removes memory entirely, so that the action depends only on the current observation and the previous action. The agents are the blind agent, the two sonar-image agents of Experiment 1 and the range + learned-beam agent of Experiment 2.

**Result 11: only the blind agents lose coverage when memory older than 20 steps is erased.** Their coverage falls from $21.3 \pm 3.2$ to $18.2 \pm 2.4$ cells ($-15\%$), and it falls in every seed ($16.8 \to 15.6$, $23.2 \to 17.5$, $23.8 \to 21.3$). For the three echo agents coverage does not fall in the mean (sonar, fixed click: $25.4 \to 27.5$; sonar, learned beam: $23.3 \to 23.9$; range, learned beam: $21.8 \to 24.2$) and collisions change by less than their standard deviation over seeds. Taken with Result 9, this supports the reading that blind explorers rely on a longer record of their own path, whereas agents that hear the room can re-derive what they need from the next echo.

**Result 12: without any memory, echo agents still cover ground but collide three to six times more; blind agents stop.** With the state erased at every step the sonar agent with a fixed click still covers $20.9 \pm 2.4$ cells (82% of its normal coverage), the sonar agent with a learned beam $16.3 \pm 2.8$ (70%), and the range + learned-beam agent $22.8 \pm 1.8$ (no loss), but their collisions rise from 22.7 to 71.4, from 15.1 to 89.3 and from 16.9 to 78.6 per episode. Short-term memory is therefore what keeps these agents from walking into the same obstacle repeatedly, while their choice of where to go is largely reactive to the current

echo. The memoryless blind agent moves 1.0 m, collides once and makes no further progress (3.2 cells in all three seeds): its only inputs are then one bump bit and its previous action.

**Repeatability of the evaluation.** The "memory intact" column repeats the evaluation of Table 4 on the same start poses. For the blind agents, which use no acoustics, the two evaluations agree exactly (per-seed coverage 16.8, 23.2, 23.8 in both). For the sonar agent with a fixed click the three-seed mean is 25.4 here against 26.2 there, and for the range + learned-beam agent 21.8 against 22.1: the reason is that Steam Audio draws new random ray directions for every simulation run, whatever the number of threads. Two clicks emitted at the same pose with identical settings give sonar images whose channel $I_0$ correlates at 0.94 and 0.97 on average in two buildings (20 poses each; the strongest cell is the same in all 40 pairs) and range profiles that correlate at 0.999 (Appendix G). The echo observations are thus noisy at a fixed pose, evaluations are not bit-reproducible, and differences of up to about one cell in a three-seed mean should not be interpreted.

## 10 Discussion

**Why echoes helped exploration but not goal reaching.** With a goal vector the agent already knows where to go; what it lacks is knowledge of obstacles on the way, and in small apartments a bump sensor plus the goal vector suffices to route around them at the cost of a few collisions (3.5 per episode for the blind agent in Table 2). In exploration the agent must decide where unvisited open space is, and echoes carry exactly that: the sonar image shows the distance to walls in every direction. This suggests that benchmarks intended to measure the value of echolocation should not provide a goal vector.

**Representation over raw signal.** The head-scan agent received strictly more acoustic simulation per step than the sonar agent and performed no better than a blind agent, whereas a deterministic transformation of one first-order Ambisonic response into range and bearing made echoes useful within 2 million steps. We do not claim that binaural input cannot work, only that at this training budget and with this encoder it did not, while the cheaper explicit representation did. For a physical device the sonar image needs a four-capsule (tetrahedral) microphone and a pulse-compressed emitter instead of two ears; its main real-world difficulty, which our simulator does not reproduce, is that arrivals overlapping within one sample blur the intensity-vector direction.

**Active sensing.** When aiming was necessary for direction (Experiment 2), agents did learn to aim and, in one seed out of three, to stay silent most of the time, but this bought safety and sensing economy, not coverage. A plausible reason is that the coverage reward favours continuous forward motion, for which a fixed forward-looking beam is sufficient, while looking sideways mainly helps to avoid side collisions.

**Memory.** The probe results show a division of labour between perception and memory that depends on the sensor: agents that sense walls keep geometry in the observation and a moderate path record in memory; agents that do not sense walls keep a stronger path record. Experiment 4 shows that this difference is functional: only the blind agents lose coverage when the record is truncated to 20 steps, and the echo agents' choice of direction survives even complete removal of memory, at the price of many more collisions.

## 11 Limitations

- **Simulation only, geometric acoustics.** The engine has no diffraction, a single material for all surfaces in our setup, three frequency bands, and the $\cos\theta$ deficit of Section 3.5. The sparse reconstruction delivers one arrival per sample, which makes the intensity-vector direction cleaner than in real recordings. No claim about real-world performance follows from these results.

- **Three seeds.** Standard deviations over three seeds are imprecise. We make claims only where all seeds of one agent exceed all seeds of another (sonar image against blind coverage; blind against echo-agent path memory) or state explicitly that a difference is within seed noise.

- **Small buildings, one dataset.** The held-out buildings have 12.5–39.6 $\mathrm{m}^2$ of navigable floor; three of the four are single rooms. Larger multi-room buildings may change the ranking, in particular for path memory.
- **Training budget.** Sonar agents were still improving at 2 million steps (Figure 5); the head-scan agents may need more.
- **Confounds we did not remove.** The blind agents of Experiment 1 were trained without the auxiliary wall-distance loss that the echo agents had. The learned-beam agent of Experiment 1 used figure-of-eight beams. The comparison between collision penalties in P5 also differs in blank-echo augmentation.
- **Probe.** The observation baseline of Experiment 3 is a pooled copy of the observation, not the full observation, and the decoders are small; AUROC values are lower bounds on what is decodable.
- **Greedy evaluation.** Sampled-action results (Appendix C) show the same ranking with smaller gaps for coverage and more collisions.

## 12 Conclusion

A 172-line patch turns Steam Audio into a simulator in which single echoes arrive at the correct sample, and makes it possible to study learned echolocation in real building geometry at several hundred environment steps per second on one desktop CPU. In that setting, echoes of the agent's own clicks did not help reach a goal whose position was given, but increased the floor area explored in unseen buildings by half when presented as a range-by-bearing image, and not at all when presented as binaural spectrograms. Aiming the click was learnable but made agents careful rather than fast. Probing and memory erasure showed that all explorers remember their own path, that blind explorers depend on that memory most, and that echo-guided explorers need memory mainly to avoid repeating collisions.

### Reproducibility statement

The repository accompanying this submission contains the patch against Steam Audio 4.8.1 and build scripts, the C validation harness and its analysis, the Python binding, the environment, training and evaluation code, the shell scripts with the exact command line of every run, the evaluation logs and per-update training logs of every run reported here, and the script that regenerates all figures and the per-building table from those logs. Policy checkpoints (about 30 MB each) and raw impulse responses are not included in the repository because of their size. All experiments ran on one workstation with an AMD Ryzen 7 5800X (8 cores), 94 GB of memory and one NVIDIA RTX 2080 Ti; the acoustics run on the CPU and the GPU is used for the policy only. The 24 training runs of Experiments 1 and 2 took 30.7 hours in total; the nine runs of the click-cost sweep took a further 9.2 hours.

### Use of AI assistance

The simulator patch, environment, training and analysis code, the experiment scripts and the first draft of this manuscript were written with an AI coding assistant working under the direction of the author, who posed the research questions, selected the experiments and takes responsibility for all content. Every number in the paper was produced by the released scripts from the released logs.

## References

Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.

Jont B. Allen and David A. Berkley. Image method for efficiently simulating small-room acoustics. *The Journal of the Acoustical Society of America*, 65(4):943–950, 1979.

Peter Anderson, Angel Chang, Devendra Singh Chaplot, Alexey Dosovitskiy, Saurabh Gupta, Vladlen Koltun, Jana Kosecka, Jitendra Malik, Roozbeh Mottaghi, Manolis Savva, and Amir R. Zamir. On evaluation of embodied navigation agents. *arXiv preprint arXiv:1807.06757*, 2018.

Angel Chang, Angela Dai, Thomas Funkhouser, Maciej Halber, Matthias Nießner, Manolis Savva, Shuran Song, Andy Zeng, and Yinda Zhang. Matterport3D: Learning from RGB-D data in indoor environments. In *International Conference on 3D Vision (3DV)*, 2017.

Devendra Singh Chaplot, Dhiraj Gandhi, Saurabh Gupta, Abhinav Gupta, and Ruslan Salakhutdinov. Learning to explore using active neural SLAM. In *International Conference on Learning Representations (ICLR)*, 2020.

Changan Chen, Unnat Jain, Carl Schissler, Sebastià Vicenç Amengual Garí, Ziad Al-Halah, Vamsi Krishna Ithapu, Philip Robinson, and Kristen Grauman. SoundSpaces: Audio-visual navigation in 3D environments. In *European Conference on Computer Vision (ECCV)*, 2020.

Changan Chen, Sagnik Majumder, Ziad Al-Halah, Ruohan Gao, Santhosh Kumar Ramakrishnan, and Kristen Grauman. Learning to set waypoints for audio-visual navigation. In *International Conference on Learning Representations (ICLR)*, 2021.

Changan Chen, Carl Schissler, Sanchit Garg, Philip Kobernik, Alexander Clegg, Paul Calamia, Dhruv Batra, Philip W. Robinson, and Kristen Grauman. SoundSpaces 2.0: A simulation platform for visual-acoustic learning. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.

Tao Chen, Saurabh Gupta, and Abhinav Gupta. Learning exploration policies for navigation. In *International Conference on Learning Representations (ICLR)*, 2019.

Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014.

Jesper Haahr Christensen, Sascha Hornauer, and Stella X. Yu. BatVision: Learning to see 3D spatial layout with two ears. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.

Itamar Eliakim, Zahi Cohen, Gabor Kosa, and Yossi Yovel. A fully autonomous terrestrial bat-like acoustic robot. *PLoS Computational Biology*, 14(9):e1006406, 2018.

Chuang Gan, Yiwei Zhang, Jiajun Wu, Boqing Gong, and Joshua B. Tenenbaum. Look, listen, and act: Towards audio-visual embodied navigation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2020.

Ruohan Gao, Changan Chen, Ziad Al-Halah, Carl Schissler, and Kristen Grauman. VisualEchoes: Spatial image representation learning through echolocation. In *European Conference on Computer Vision (ECCV)*, 2020.

Donald R. Griffin. *Listening in the Dark: The Acoustic Orientation of Bats and Men*. Yale University Press, 1958.

Xixi Hu, Senthil Purushwalkam, David Harwath, and Kristen Grauman. Learning to map efficiently by active echolocation. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

Max Jaderberg, Volodymyr Mnih, Wojciech Marian Czarnecki, Tom Schaul, Joel Z. Leibo, David Silver, and Koray Kavukcuoglu. Reinforcement learning with unsupervised auxiliary tasks. In *International Conference on Learning Representations (ICLR)*, 2017.

Juha Merimaa and Ville Pulkki. Spatial impulse response rendering I: Analysis and synthesis. *Journal of the Audio Engineering Society*, 53(12):1115–1127, 2005.

Piotr Mirowski, Razvan Pascanu, Fabio Viola, Hubert Soyer, Andrew J. Ballard, Andrea Banino, Misha Denil, Ross Goroshin, Laurent Sifre, Koray Kavukcuoglu, Dharshan Kumaran, and Raia Hadsell. Learning to navigate in complex environments. In *International Conference on Learning Representations (ICLR)*, 2017.

Adithya Venkatesh Mohan and Dieter Vanderelst. Training a simulated bat: Modeling sonar-based obstacle avoidance using deep-reinforcement learning. In *IEEE Symposium Series on Computational Intelligence (SSCI)*, pp. 2241–2249, 2020.

Cynthia F. Moss and Annemarie Surlykke. Probing the natural scene by echolocation in bats. *Frontiers in Behavioral Neuroscience*, 4:33, 2010.

Thinh H. Nguyen and Dieter Vanderelst. Toward behavior-based models of bat echolocation. In *IEEE Symposium Series on Computational Intelligence (SSCI)*, 2022.

Ville Pulkki. Spatial sound reproduction with directional audio coding. *Journal of the Audio Engineering Society*, 55(6):503–516, 2007.

Senthil Purushwalkam, Sebastià Vicenç Amengual Garí, Vamsi Krishna Ithapu, Carl Schissler, Philip Robinson, Abhinav Gupta, and Kristen Grauman. Audio-visual floorplan reconstruction. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain, Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, Devi Parikh, and Dhruv Batra. Habitat: A platform for embodied AI research. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations (ICLR)*, 2016.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Julian Straub, Thomas Whelan, Lingni Ma, Yufan Chen, Erik Wijmans, Simon Green, Jakob J. Engel, Raul Mur-Artal, Carl Ren, Shobhit Verma, Anton Clarkson, Mingfei Yan, Brian Budge, Yajie Yan, Xiaqing Pan, June Yon, Yuyang Zou, Kimberly Leon, Nigel Carter, Jesus Briales, Tyler Gillingham, Elias Mueggler, Luis Pesqueira, Manolis Savva, Dhruv Batra, Hauke M. Strasdat, Renzo De Nardi, Michael Goesele, Steven Lovegrove, and Richard Newcombe. The Replica dataset: A digital replica of indoor spaces. *arXiv preprint arXiv:1906.05797*, 2019.

Lore Thaler and Melvyn A. Goodale. Echolocation in humans: an overview. *Wiley Interdisciplinary Reviews: Cognitive Science*, 7(6):382–393, 2016.

Valve Corporation. Steam Audio, version 4.8.1. `https://github.com/ValveSoftware/steam-audio`, 2026.

Ingo Wald, Sven Woop, Carsten Benthin, Gregory S. Johnson, and Manfred Ernst. Embree: a kernel framework for efficient CPU ray tracing. *ACM Transactions on Graphics*, 33(4):143:1–143:8, 2014.

Erik Wijmans, Manolis Savva, Irfan Essa, Stefan Lee, Ari S. Morcos, and Dhruv Batra. Emergence of maps in the memories of blind navigation agents. In *International Conference on Learning Representations (ICLR)*, 2023.

Yinfeng Yu, Lele Cao, Fuchun Sun, Chao Yang, Huicheng Lai, and Wenbing Huang. Echo-enhanced embodied visual navigation. *Neural Computation*, 35(5):958–976, 2023.

Lingyu Zhu, Esa Rahtu, and Hang Zhao. Beyond visual field of view: Perceiving 3D environment with echoes and vision. *arXiv preprint arXiv:2207.01136*, 2022.

## A Simulator validation in full

All single-wall tests use a C program that calls the Steam Audio C API directly: one planar wall (200 m × 200 m unless stated), one source, one listener, one bounce, first-order Ambisonics, 48 kHz, 65,536 rays for the stock library and 262,144 rays for the patched library. The impulse response is obtained by sending a unit impulse through the engine's convolution reflection effect after one silent frame. About 100 scenarios are run per library and compared with closed-form predictions by a script that writes a pass/fail list. Table 8 lists the checks on the unmodified library and Table 9 those on the patched library.

Table 8: Checks on the unmodified Steam Audio 4.8.1 library (25 of 25 pass).

| Quantity | Result |
|---|---|
| Reflection delay | reflection energy appears in the correct 10 ms bin of $(r_{\text{refl}} - r_{\text{direct}})/c$ for 13 wall distances and 5 source distances |
| Speed of sound | bin indices at five path lengths chosen to separate the two hypotheses agree with 340 m/s in 5 of 5 cases and with 343 m/s in 2 of 5 |
| Absorption | $E(\alpha)/E(0) = 1 - \alpha$ to four decimals; per-band spectral ratios within 0.02 of the band-filter model |
| Distance | $E \propto r_{\text{refl}}^{-2}$, log–log slope −1.998, spread 0.24% over 6–103 m; energy follows the image-source path length, not the source-to-wall distance |
| Absolute level | measured 5.60 against 5.93 predicted from the noise reconstruction constants |
| Direction | first-order channels point to the image source; N3D gain 1.715 (expected 1.732) |
| Scattering | on a 2×2 m panel $E(s)/E(0) = 1 + s(R - 1)$ with $R = 0.453$ (Lambertian/specular ratio), within 0.015 |
| Finite panel | specular point on the panel: 0.998 of the infinite wall; specular point 5 m off the panel: $4 \times 10^{-12}$ |
| Direct path | occlusion 0 behind the wall, 1 otherwise; attenuation $1/\max(r, 1\,\text{m})$; air absorption $e^{-k_b r}$ exact |
| Geometry | reversed triangle winding gives an identical reflection; a source behind the wall gives none; an empty scene gives a zero response |
| Ray count | 65,536 rays within $10^{-4}$ of 262,144 rays in energy; 1,024 rays within 4% |

Table 9: Checks on the patched library with one-sample bins and sparse reconstruction (26 of 26 pass).

| Quantity | Result |
|---|---|
| Arrival time | onset sample equals $\lfloor (r_{\text{refl}} - 1\,\text{cm} - r_{\text{direct}}) f_s/c \rfloor$ within 1 sample for 13 wall distances from 1 to 15 m; fitted speed of sound 340.01 m/s |
| Long range | 34–102 m paths: onset within 0.8 sample with $10^6$ rays; with 65,536 rays only 1–2 rays fall in each sample-wide ring of the specular spot and onsets jitter by a few samples |
| Glossy tail | centroid minus onset is 0.87–0.98 of the predicted mean excess path $2w/(n+1)$ for wall distances $w$ = 1.5–40 m |
| Distance | $E\,r^2$ constant to 0.3% over 3–80 m (source at listener) and 0.26% over 6–103 m (source apart); slopes −2.003 and −1.998 |
| Absolute level | $E \,/\, [(1 + 2/n)/(4\pi r^2)] = 12.2$ against $4\pi = 12.57$ |
| Absorption | $1 - \alpha$ to four decimals; band ratios within 0.024 of the complementary-filter model |
| Air absorption | per band within 0.018 of $e^{-k_b d}$ evaluated at each sample's distance |
| Scattering | infinite diffuse wall: energy ratio 2.02 against 1.96 analytic; 2×2 m panel within 0.016 |
| Direction, occlusion, transmission, two-sidedness, empty scene | same outcome as for the stock library |

**Shoebox room.** 6×4×3 m, source and listener at different positions, 262,144 rays, patched library. (1) All six first-order echoes begin at the sample predicted by the image-source model. (2) Their energies are 0.92–1.15 of the image-source prediction once the $\cos\theta$ factor of Section 3.5 is applied at the last wall, and 0.70–1.04 of a pure mirror model. (3) Their energy-weighted direction vectors differ from the predicted

direction vectors by at most 0.15. (4) The energy decay over the first 40 ms is within 1.05 dB of a three-bounce image-source model with the $\cos\theta$ factor. (5) Reverberation time by Schroeder backward integration, divided by the Eyring prediction, is 0.91, 1.04, 1.11 for diffuse walls with $\alpha = 0.1, 0.3, 0.5$ and 1.05, 1.17, 1.24 for specular walls. (6) Steam Audio's own parametric reverberation estimate agrees with the Schroeder value of its impulse response to 1%. (7) The Embree build gives the same ratios. Seven of seven checks pass.

**Echo timing inside the learning environment.** In a Habitat test apartment (382,000 triangles), at 12 random poses, the first echo in the binaural click response arrived at $2d/c$ for the nearest surface distance $d$ with a maximum error of 0.7 ms after removing the 3.25 ms latency of the binaural decoder.

## B Buildings

Table 10 lists the navigable floor area of every Replica scene as reported by Habitat's navigation mesh for an agent of radius 0.1 m and height 1.5 m.

Table 10: Replica scenes and navigable area. One training environment runs in each of the 14 training scenes.

| Training scene | $m^2$ | Training scene | $m^2$ | Held-out scene | $m^2$ |
|---|---|---|---|---|---|
| apartment_0 | 89.0 | frl_apartment_5 | 44.2 | office_2 | 12.5 |
| apartment_1 | 52.9 | office_0 | 7.6 | frl_apartment_4 | 39.6 |
| apartment_2 | 44.7 | office_1 | 5.4 | hotel_0 | 16.0 |
| frl_apartment_0 | 40.4 | office_3 | 20.3 | room_2 | 13.2 |
| frl_apartment_1 | 42.0 | office_4 | 22.9 | | |
| frl_apartment_2 | 39.2 | room_0 | 17.1 | | |
| frl_apartment_3 | 42.3 | room_1 | 11.1 | | |

## C Per-building and sampled-action results

Table 11 gives the greedy-policy results of every seed in every held-out building for Experiments 1 and 2. Table 12 repeats Tables 3 and 5 for actions sampled from the policy distribution.

## D Probe results with linear decoders

Table 13 gives all probe results of Experiment 3, including linear decoders and the *visited* label evaluated over all cells (which, as explained in Section 8, can be partly inferred from geometry).

## E Details of the preliminary single-seed runs

**P2 in numbers.** Point-goal in the dark in the Habitat test scene `apartment_1`, 100 held-out episodes, sampled actions. After 1.5 million steps with $\lambda = 0.1$: head-scan echo agent, success 0.64, SPL 0.49, 12.2 collisions per episode; single-orientation echo agent, 0.44, 0.38, 9.7; blind agent, 0.20, 0.19, 1.8 (the blind agent's low collision count reflects that it stops after 11–21 steps). With shuffled echoes the two echo agents fall to 0.36 and 0.27 success with 50 and 19.5 collisions. After 0.6 million steps, zeroing the echoes changed collisions per episode from 2.7 to 22 and from 8.0 to 24 (single orientation, seeds 1 and 2) and from 2.4 to 53 and from 11.5 to 40 (head scan, seeds 1 and 2).

**P3 in numbers.** Training on `apartment_1`, `van-gogh-room` and `skokloster-castle` for 2 million steps; point-goal success (greedy) of the head-scan agent with auxiliary loss against the blind agent: 0.35 against 0.29, 0.93 against 0.85, 0.88 against 0.73 in the three training scenes, and 0.18 against 0.21 in the unseen Matterport3D house `17DRP5sb8fy` (0.31 with shuffled echoes). For exploration in the same unseen house the head-scan explorer covered 12.6 cells with real echoes, 20.4 with shuffled echoes and 29.1 with zeroed echoes,

Table 11: Greedy-policy results per held-out building and seed (30 episodes each).

| | | Coverage (cells) | | | | Collisions per episode | | | |
|---|---|---|---|---|---|---|---|---|---|
| Agent | Building | seed 1 | seed 2 | seed 3 | mean | seed 1 | seed 2 | seed 3 | mean |
| Sonar image, fixed click | office_2 | 24.4 | 24.5 | 27.9 | 25.6 | 19.0 | 16.8 | 40.1 | 25.3 |
| | frl_apartment_4 | 58.9 | 51.6 | 56.8 | 55.8 | 28.7 | 9.2 | 22.6 | 20.2 |
| | hotel_0 | 29.0 | 26.7 | 25.0 | 26.9 | 18.0 | 7.0 | 25.9 | 17.0 |
| | room_2 | 43.3 | 37.8 | 38.7 | 39.9 | 10.2 | 2.1 | 10.9 | 7.7 |
| Sonar image + learned beam | office_2 | 27.0 | 23.6 | 19.5 | 23.4 | 23.4 | 13.6 | 9.1 | 15.4 |
| | frl_apartment_4 | 52.3 | 63.5 | 47.7 | 54.5 | 14.5 | 11.7 | 9.5 | 11.9 |
| | hotel_0 | 27.5 | 23.2 | 19.6 | 23.4 | 19.3 | 10.1 | 16.2 | 15.2 |
| | room_2 | 43.7 | 33.9 | 21.3 | 33.0 | 9.1 | 9.2 | 13.3 | 10.5 |
| Head-scan binaural | office_2 | 10.5 | 17.8 | 19.2 | 15.8 | 12.2 | 20.2 | 27.8 | 20.1 |
| | frl_apartment_4 | 43.0 | 37.1 | 37.9 | 39.3 | 21.1 | 19.5 | 25.2 | 21.9 |
| | hotel_0 | 16.7 | 21.1 | 18.8 | 18.9 | 10.3 | 16.7 | 28.6 | 18.5 |
| | room_2 | 17.4 | 40.1 | 39.3 | 32.3 | 9.0 | 12.3 | 5.8 | 9.0 |
| Blind | office_2 | 17.8 | 24.8 | 26.3 | 23.0 | 14.0 | 27.5 | 25.4 | 22.3 |
| | frl_apartment_4 | 23.6 | 32.9 | 32.1 | 29.5 | 12.7 | 21.1 | 22.5 | 18.8 |
| | hotel_0 | 15.8 | 21.7 | 21.3 | 19.6 | 13.2 | 26.3 | 27.1 | 22.2 |
| | room_2 | 18.8 | 28.4 | 31.0 | 26.1 | 12.6 | 20.7 | 23.8 | 19.0 |
| Range + learned beam | office_2 | 23.5 | 26.4 | 25.3 | 25.1 | 15.3 | 21.5 | 18.9 | 18.6 |
| | frl_apartment_4 | 34.8 | 36.8 | 28.5 | 33.4 | 16.5 | 14.9 | 14.5 | 15.3 |
| | hotel_0 | 18.5 | 20.3 | 18.7 | 19.2 | 9.7 | 20.5 | 14.3 | 14.8 |
| | room_2 | 31.7 | 34.7 | 32.9 | 33.1 | 10.7 | 15.2 | 13.1 | 13.0 |
| Range + scripted sweep | office_2 | 24.9 | 26.2 | 27.5 | 26.2 | 21.3 | 30.7 | 36.4 | 29.5 |
| | frl_apartment_4 | 35.5 | 34.2 | 37.5 | 35.7 | 27.2 | 21.3 | 34.7 | 27.7 |
| | hotel_0 | 23.7 | 22.2 | 23.1 | 23.0 | 23.0 | 19.1 | 22.9 | 21.7 |
| | room_2 | 35.7 | 39.7 | 34.7 | 36.7 | 16.6 | 17.8 | 22.2 | 18.9 |
| Range + fixed ahead beam | office_2 | 22.5 | 29.2 | 18.9 | 23.5 | 22.5 | 40.5 | 49.3 | 37.4 |
| | frl_apartment_4 | 42.4 | 43.7 | 41.5 | 42.5 | 12.0 | 46.8 | 76.2 | 45.0 |
| | hotel_0 | 20.5 | 23.4 | 15.1 | 19.7 | 11.9 | 30.9 | 15.5 | 19.4 |
| | room_2 | 44.5 | 46.5 | 36.2 | 42.4 | 7.9 | 16.0 | 22.0 | 15.3 |
| Range + omni click | office_2 | 26.4 | 25.0 | 11.8 | 21.1 | 39.3 | 43.2 | 5.1 | 29.2 |
| | frl_apartment_4 | 34.8 | 41.8 | 30.8 | 35.8 | 45.6 | 43.6 | 16.8 | 35.3 |
| | hotel_0 | 24.0 | 26.3 | 15.5 | 21.9 | 44.1 | 43.2 | 10.6 | 32.6 |
| | room_2 | 48.2 | 47.0 | 10.3 | 35.2 | 26.4 | 30.8 | 3.3 | 20.2 |

Table 12: Results with sampled actions on the four held-out buildings (mean ± s.d. over 3 seeds).

| Agent | Coverage (cells) | Collisions / episode |
|---|---|---|
| Sonar image, fixed click | $38.4 \pm 0.8$ | $26.4 \pm 7.4$ |
| Sonar image + learned beam | $36.6 \pm 2.2$ | $24.0 \pm 4.0$ |
| Head-scan binaural | $31.1 \pm 1.9$ | $32.5 \pm 5.1$ |
| Blind | $27.8 \pm 1.4$ | $31.3 \pm 4.6$ |
| Range + fixed ahead beam | $33.3 \pm 1.1$ | $42.1 \pm 12.2$ |
| Range + scripted sweep | $31.8 \pm 0.7$ | $37.6 \pm 4.5$ |
| Range + omni click | $31.2 \pm 3.4$ | $42.9 \pm 15.7$ |
| Range + learned beam | $30.9 \pm 1.6$ | $27.3 \pm 5.7$ |

against 26.3 for the blind agent and 32.6 for bump-and-turn: with too few training buildings, real echoes in an unfamiliar building suppressed movement.

**Exploration with $\lambda = 0.02$ (one seed per agent).** Held-out coverage / collisions: sonar image with fixed click 41.9 / 38.6; head-scan binaural 36.7 / 18.6; sonar image with learned beam 25.6 / 8.6; blind 18.2 / 12.9; bump-and-turn 28.5 / 31.5. These runs had no blank-echo augmentation. The head-scan result at

Table 13: Experiment 3, held-out AUROC (mean ± s.d. over 3 seeds). "State" = decoder input is the GRU state; "Obs." = decoder input is the pooled current observation.

| | | Walkable | | Visited, all cells | | Visited, walkable cells | |
|---|---|---|---|---|---|---|---|
| Agent | Decoder | State | Obs. | State | Obs. | State | Obs. |
| Sonar, fixed click | linear | $0.790 \pm 0.008$ | $0.806 \pm 0.006$ | $0.755 \pm 0.021$ | $0.781 \pm 0.008$ | $0.677 \pm 0.022$ | $0.618 \pm 0.008$ |
| | MLP | $0.803 \pm 0.007$ | $0.832 \pm 0.002$ | $0.861 \pm 0.003$ | $0.808 \pm 0.007$ | $0.762 \pm 0.006$ | $0.625 \pm 0.009$ |
| Sonar + learned beam | linear | $0.775 \pm 0.014$ | $0.749 \pm 0.017$ | $0.739 \pm 0.024$ | $0.714 \pm 0.017$ | $0.674 \pm 0.015$ | $0.582 \pm 0.007$ |
| | MLP | $0.789 \pm 0.016$ | $0.779 \pm 0.019$ | $0.840 \pm 0.006$ | $0.735 \pm 0.019$ | $0.755 \pm 0.010$ | $0.579 \pm 0.005$ |
| Head-scan binaural | linear | $0.754 \pm 0.014$ | $0.745 \pm 0.015$ | $0.780 \pm 0.032$ | $0.722 \pm 0.014$ | $0.708 \pm 0.023$ | $0.575 \pm 0.015$ |
| | MLP | $0.755 \pm 0.016$ | $0.761 \pm 0.018$ | $0.831 \pm 0.012$ | $0.747 \pm 0.013$ | $0.757 \pm 0.009$ | $0.594 \pm 0.011$ |
| Blind | linear | $0.611 \pm 0.006$ | $0.502 \pm 0.002$ | $0.774 \pm 0.046$ | $0.509 \pm 0.010$ | $0.763 \pm 0.043$ | $0.513 \pm 0.013$ |
| | MLP | $0.614 \pm 0.002$ | $0.503 \pm 0.003$ | $0.843 \pm 0.010$ | $0.511 \pm 0.010$ | $0.830 \pm 0.011$ | $0.515 \pm 0.013$ |

this penalty (36.7 cells, one seed) is higher than its three-seed mean at $\lambda = 0.05$ (26.6); we cannot separate the effect of the penalty from seed variation with one run.

## F Hyperparameters

Table 14: Settings shared by all runs of Experiments 1 and 2.

| Acoustic simulation | | PPO | |
|---|---|---|---|
| Rays per click | 2,048 | Parallel environments | 14 (one per training scene) |
| Bounces | 4 | Rollout length | 128 steps |
| Response length | 0.3 s (observation uses 0.128 s) | Epochs / minibatches | 4 / 4 |
| Sampling rate | 24 kHz | Learning rate (Adam, $\epsilon = 10^{-5}$) | $2.5 \times 10^{-4}$ |
| Ambisonic order | 1 | Discount $\gamma$ / GAE $\lambda$ | 0.99 / 0.95 |
| Bin duration | 1/24000 s, sparse reconstruction | Clip range | 0.2 |
| Absorption (3 bands) | 0.10, 0.20, 0.30 | Entropy / value coefficient | 0.02 / 0.5 |
| Scattering | 0.05 | Gradient-norm clip | 0.5 |
| Minimum irradiance | 0.1 | Total steps | 2,001,664 |
| Ray tracer | Embree, 1 thread per environment | Auxiliary loss weight | 0.5 (echo agents) |
| Speed of sound | 340 m/s | Blank-echo probability | 0.05 |
| **Environment** | | **Rewards (exploration)** | |
| Forward step / turn | 0.25 m / 10° | New 0.5 m cell | +1 |
| Agent radius / ear height | 0.1 m / 1.5 m | Per step | −0.01 |
| Wall sliding | off | Per collided step | −0.05 |
| Episode length | 300 steps | Per click (learned-beam agents) | −0.01 |

## G Additional measurements

**Click-to-click variability.** With one thread and the same simulator object, two consecutive simulations of the same click (`office_2`, learning settings) differ by up to 0.16 in impulse-response amplitude (peak 1.35) and by 1.2% in total energy; 1,454 of the 4,096 sonar-image cells change. With eight threads the figures are 0.11, 0.9% and 1,368 cells, so threading is not the cause. Over 20 poses per building the correlation between the $I_0$ channels of two identical clicks is 0.942 (minimum 0.913) in `office_2` and 0.965 (minimum 0.924) in `frl_apartment_4`; with 16,384 rays instead of 2,048 it is 0.925 and 0.959, i.e. not higher. Range profiles correlate at 0.999 (minimum 0.997).

**Where and when path memory is decodable.** Table 15 breaks the MLP state probes of Experiment 3 down by distance from the agent and by time within the episode. Path memory is equally decodable near and far from the agent and ahead of and behind it, is highest for the cell directly behind the agent, and declines as the episode proceeds (for the sonar agent from 0.91 in the first 50 steps to 0.67 after step 150; for the blind agent from 0.95 to 0.74). The blind agents' knowledge of walkable space is local: 0.75 within 1 m and 0.58 beyond 2 m.

Table 15: Experiment 3 in more detail: held-out AUROC of the MLP decoder on the recurrent state (mean over 3 seeds). "Near", "mid" and "far" are cells whose centres lie within 1 m, 1–2 m and more than 2 m of the agent.

| | Walkable | | | Path memory | | | Path memory by episode step | | |
|---|---|---|---|---|---|---|---|---|---|
| Agent | near | mid | far | near | mid | far | 0–49 | 50–149 | 150–299 |
| Sonar image, fixed click | 0.810 | 0.782 | 0.813 | 0.782 | 0.770 | 0.754 | 0.910 | 0.739 | 0.666 |
| Sonar image + learned beam | 0.795 | 0.768 | 0.799 | 0.762 | 0.750 | 0.757 | 0.883 | 0.724 | 0.657 |
| Head-scan binaural | 0.765 | 0.726 | 0.768 | 0.768 | 0.756 | 0.756 | 0.884 | 0.734 | 0.673 |
| Blind | 0.752 | 0.633 | 0.582 | 0.824 | 0.828 | 0.832 | 0.951 | 0.849 | 0.739 |